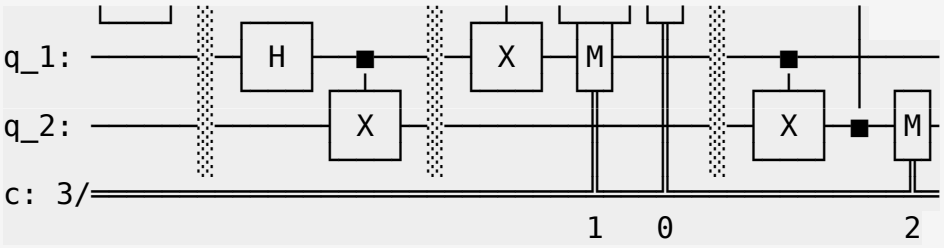
Practical 2

```python
from qiskit import QuantumCircuit, transpile
from qiskit_aer import AerSimulator

qc = QuantumCircuit(3, 3)

# State to teleport
qc.x(0)

qc.barrier()

# Create Bell pair
qc.h(1)
qc.cx(1, 2)

qc.barrier()

# Teleportation
qc.cx(0, 1)
qc.h(0)

qc.measure([0, 1], [0, 1])

qc.barrier()

# Correction operations
qc.cx(1, 2)
qc.cz(0, 2)

qc.measure(2, 2)

simulator = AerSimulator()
compiled = transpile(qc, simulator)

result = simulator.run(compiled, shots=1000).result()

print("Counts:")
print(result.get_counts())

print("\nCircuit:")
print(qc.draw())
```

```
Counts:
{'100': 252, '111': 226, '110': 261, '101': 261}

Circuit:
```


q_1:
q_2:
c: 3/
H
X
X
M
X
M
1
0
2